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Project Title:	Computational Models of Cardiovascular Function for Simulation, Data Integration, and Clinical Decision Support		
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Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Cardiovascular Alterations Team		
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Contact Monitor:		**Contact Phone:**	
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COI Name (Institution):	Heldt, Thomas (Massachusetts Institute of Technology) Verghese, George (Massachusetts Institute of Technology)		
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	One of the highest priority problems in the current manned space program is orthostatic intolerance (OI) experienced by astronauts upon their return to the normal gravitational environment. This problem has been well known since the earliest days of manned space flight, and has been intensely studied in-flight and in many ground-based (bed-rest) studies. A number of countermeasures have been proposed and evaluated, but to date no effective and practical countermeasure has been developed. While the effectiveness of countermeasures can only be explored in ground-based experiments and ultimately be proven successful when used by astronauts, the small number of subjects studied in both situations makes the interpretation of limited experimental data for countermeasure development and refinement difficult. Mathematical models that represent the essential functional components of the cardiovascular system provide a powerful		

Task Description:	means to extract physiologically relevant information from multiparameter experimental data. We will extend progress already made, with the following specific aims: 1. Develop procedures and algorithms for automated model-based extraction of physiologically relevant information from multivariate data streams, particularly those relevant to countermeasure development for OI. We will expand our unified data archive of hemodynamic signals during gravitational stress. 2. Develop strategies for systematic and effective model identification, i.e., for the adaptation of model structures and parameters to the characteristics of the available physiological data. These strategies will be based on analytical and computational studies of candidate models and their sensitivities to parameter changes. 3. Apply our computational modeling and parameter identification technology to the challenge of cardiovascular monitoring of critically ill patients. Hemodynamic measurements will be transformed into physiologically meaningful parameters through model-based processing. 4. Develop and thoroughly test a graphical user interface for our current cardiovascular model. The resultant cardiovascular simulator will be open-source and capable of running on all major operating systems. Furthermore, the entire model will be made publicly available to researchers and educators. During the second year of the current funding cycle, major progress was made in the following topical areas: 1) Model refinement, exploration, and extension In collaboration with our colleagues from the Sensorimotor Adaptation Team, we have begun adapting our models to allow for simulation of short-radius centrifugation. The simulations will provide a basis for predicting cardiovascular stress as a function of human anthropometric measures and centrifuge parameters. 2) Model analysis and identification Major progress was made in analyzing and simplifying the model structure of our cardiovascular models. Using physiologically motivated approximations, we have been able to solve the short- and long-term dynamics of a minimal cardiovascular model analytically. Such analytical solutions are very valuable for parameter estimation problems; they also allow for faster simulations as they do not require numerical integration of the associated differential equations. The results of this work are extendible to larger versions of the cardiovascular model considered here. 3) Model-based tracking of patients in intensive care: We have adapted the model and identification methodologies for the context of parameter estimation from critically ill patients in intensive care units. Using simulated, noise corrupted data, our results indicate that tracking of clinically important cardiovascular parameters (distending blood volume, peripheral resistance, and end-diastolic right ventricular capacitance) is feasible. 4) Development of a platform-independent version of the model We have begun development and testing of platform-independent (JAVA-based) versions of our models. In the initial phase, major effort is devoted to the design, implementation, and testing of an intuitive graphical user interface. We intend to pursue the following goals during year 3 of the funding period: - Apply the parameter estimation algorithm to astronaut data. We will continue our collaboration with the Cardiovascular Laboratory at Johnson Space Center and offer our computational models to interpret the hemodynamic data obtained from astronauts before and after space travel. - In collaboration with our colleagues from the NSBRI Sensorimotor Adaptation Team, we will test and validate a computational model of the hemodynamic response to short-radius centrifugation. Since the orthostatic stress during centrifugation is not constant, but dependent on the radial distance from the center of rotation, predicting the cardiovascular response to a particular acceleration stress is a non-trivial task that can be greatly aided by having a reasonably validated computer model. - We will continue to identify and test optimal parameter estimation strategies for the transient hemodynamic responses recorded during orthostatic stress (tilt, LBNP, standing-up, centrifugation). - We will continue to evaluate model identification algorithms for tracking of cardiovascular parameters of the critically ill. We will investigate how our parameter estimates depend on the availabitlity of individual measurement signals. - We will continue to develop and evaluate the JAVA-based version of our model with the goal of releasing this platform-independent version of the model early in the third year of funding. The results obtained during the first two years of the current funding cycle demonstrate the utility of the models in interpreting hemodynamic data from astronauts and intensive care patients alike. In particular, these models for comparison of the relative impact deleterious changes in cardiovascular parameters can have on the ove
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Outside the scope of modeling the effects of microgravity, numerical cardiovascular simulations 1. provide a basis for interpretation of clinical data in the context of intensive care. Simulations are basic components of an intelligent patient monitoring system which can couple real-time physiologic data to cardiovascular pathology. 2. provide a powerful environment for teaching of normal and abnormal cardiovascular physiology.
	We anticipate the following progress to be made in the final year of funding: 1) Model application to orthostatic stress simulations: - In collaboration with our colleagues from Johnson Space Center, we will continue to analyze retrospectively astronaut stand test data using our model identification algorithms to identify common cardiovascular parameter changes that can be attributed to the astronauts' exposure to microgravity. - In collaboration with our colleagues from the NSBRI Sensorimotor Adaptation Team, we will test and validate a computational model of the hemodynamic response to short-radius centrifugation. Since the orthostatic stress during centrifugation is not constant, but dependent on the radial distance from the center of rotation, predicting the cardiovascular response to a particular acceleration stress is a non-trivial task that can be greatly aided by having a reasonably validated computer model.

Task Progress:	2) Model reduction and identification: - We will continue to identify and test optimal parameter estimation strategies for the transient hemodynamic responses recorded during orthostatic stress (tilt, LBNP, standing-up, centrifugation). 3) Tracking of cardiovascular parameters in the intensive care unit - We will continue to evaluate model identification algorithms for tracking of cardiovascular parameters of the critically ill. We will investigate how our parameter estimates depend on the availabitlity of individual measurement signals. 4) Developement of a platform independent model - We will continue to develop and evaluate the JAVA-based version of our model with the goal of releasing this platform-independent version of the model by the end of the third year of funding.
Bibliography Type:	Description: (Last Updated: 08/08/2019)
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